

Mixture viscosity measurement — Solution

W26 (2026) — English translation

A1

0.20 pts

Express the mole fraction of water n_w in solution in terms of the mass fraction of glycerol w_g , μ_w and μ_g .

The amount of a substance in a mixture is defined as:

$$\nu = \frac{m}{\mu},$$

where m is the mass of the substance, and μ is its molar mass.

Let us write the formula for the molar concentration of water:

$$n_w = 1 - n_g = 1 - \frac{\nu_g}{\nu_g + \nu_w},$$

from which we obtain:

Answer:

$$n_w = 1 - \frac{w_g \mu_w}{\mu_g - w_g(\mu_g - \mu_w)}$$

A2

1.50 pts

Using the equation (1), convert the values of T_g to the values of w_g .

Using the tabular method or the simple iteration method with $a = 0.9$, we obtain:

Answer:

T_g , K	w_g	η , Pa · s
195.2	1.000	0.906
194.8	0.996	0.839
194.6	0.994	0.767
193.9	0.987	0.655
193.1	0.979	0.558
191.8	0.965	0.479
190.8	0.954	0.373
188.6	0.930	0.202
183.4	0.869	0.0674
173.0	0.728	0.0278
170.0	0.682	0.0134
154.6	0.384	0.0026
149.3	0.252	0.0017
143.8	0.100	0.0012
140.3	0	0.0009

A3

0.60 pts

Propose a linearization that describes a system with a small mole fraction of water, which can be used to determine the parameters ξ and k . Plot the linearized relationship. Determine the parameters ξ and k

using the resulting graph.

Let us recalculate the data to find the dependence $\ln\left(\frac{\eta}{1 \text{ Pa} \cdot \text{s}}\right)(n_w)$.

$T_g, \text{ K}$	w_g	$\eta, \text{ Pa} \cdot \text{s}$	n_w	$\ln \eta$
195.2	1.000	0.906	0	-0.099
194.8	0.996	0.839	0.020	-0.176
194.6	0.994	0.767	0.030	-0.265
193.9	0.987	0.655	0.063	-0.423
193.1	0.979	0.558	0.099	-0.583
191.8	0.965	0.479	0.156	-0.736
190.8	0.954	0.373	0.198	-0.986
188.6	0.930	0.202	0.278	-1.600
183.4	0.869	0.0674	-	-
173.0	0.728	0.0278	-	-
170.0	0.682	0.0134	-	-
154.6	0.384	0.0026	-	-
149.3	0.252	0.0017	-	-
143.8	0.100	0.0012	-	-
140.3	0	0.0009	-	-

Let us plot the resulting dependence in the range $n_w \in [0, 0.2]$.
Approximating the dependence with a straight line, we obtain:

$$k = 4.31,$$

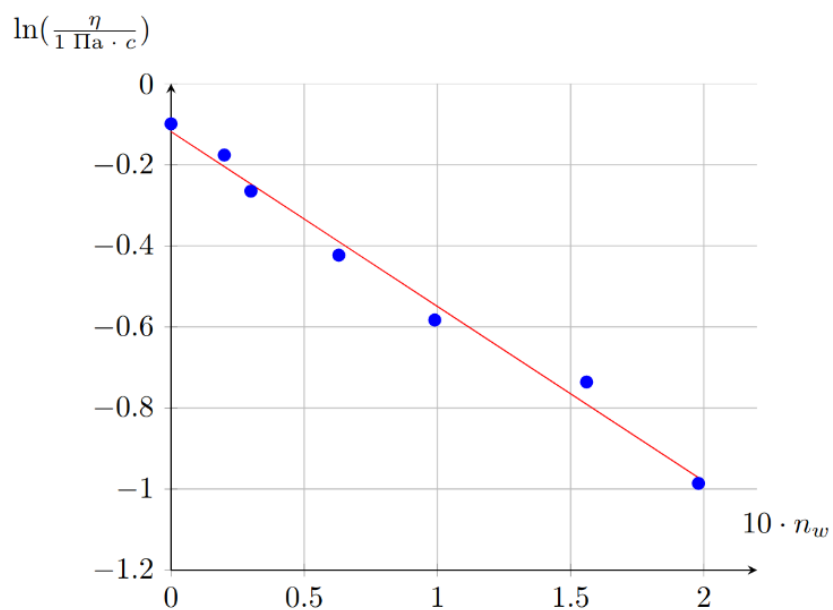
$$\ln\left(\frac{\xi}{1 \text{ Pa} \cdot \text{s}}\right) = -0.12,$$

from which it follows:

Answer:

$$k = 4.31$$

$$\xi = 0.887 \text{ Pa} \cdot \text{s}$$



Knowing the parameters ξ and k , propose a linearization that allows you to determine the values of the parameters a and b . Plot the linearized relationship. Using the straight section, determine a and b .

Let us continue recalculating the data points:

T_g , K	w_g	η , Pa · s	n_w	$\ln \left \frac{\xi \exp(-kn_w)}{\eta} - 1 \right $
195.2	1.000	0.906	0	-3.92
194.8	0.996	0.839	0.020	-3.54
194.6	0.994	0.767	0.030	-4.12
193.9	0.987	0.655	0.063	-3.44
193.1	0.979	0.558	0.099	-3.28
191.8	0.965	0.479	0.156	-2.91
190.8	0.954	0.373	0.198	-4.35
188.6	0.930	0.202	0.278	-1.12
183.4	0.869	0.0674	0.435	0.02
173.0	0.728	0.0278	0.656	-0.12
170.0	0.682	0.0134	0.704	0.78
154.6	0.384	0.0026	0.891	1.85
149.3	0.252	0.0017	0.938	2.10
143.8	0.100	0.0012	0.979	2.29
140.3	0	0.0009	1	2.50

Let us plot the resulting points on the graph $\ln \left| \frac{\xi \exp(-kn_w)}{\eta} - 1 \right| (n_w)$.

In the range $n_w \in [0, 0.2]$, the expression inside the logarithm changes sign and is close to zero, so the function behaves non-monotonically in this region. To perform a linear fit, we consider only points in the range $n_w \in [0.7, 1]$.

For $n_w > 0.2$, $\frac{\xi \exp(-kn_w)}{\eta} > 1$ holds true, so $a > 0$. By drawing the approximating straight line, we find:

$$b = 5.66,$$

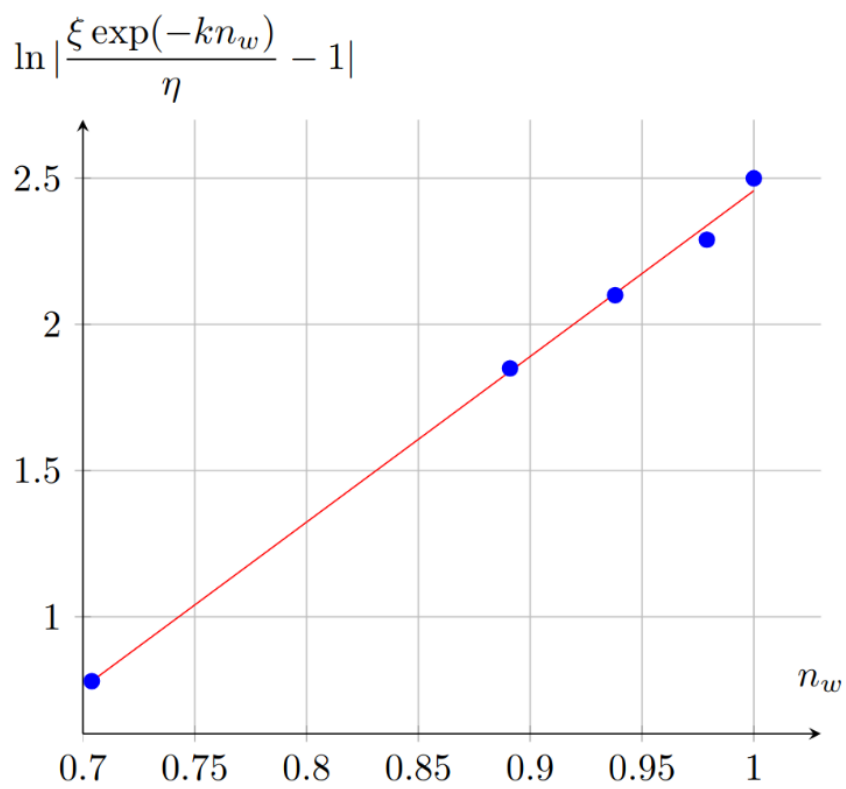
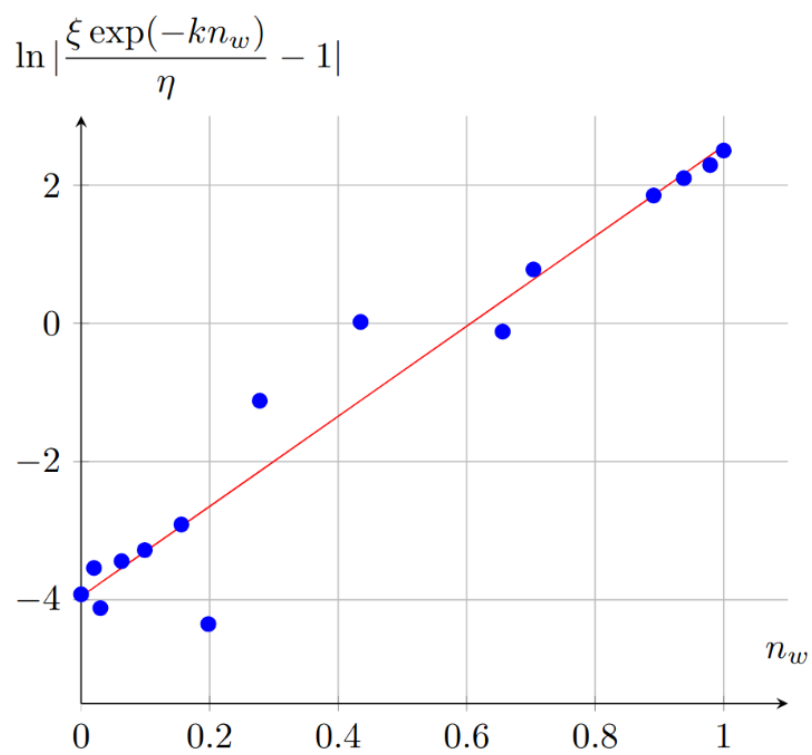
$$\ln a = -3.21,$$

from which we obtain:

Answer:

$$a = 0.040$$

$$b = 5.66$$



Draw a graph of the theoretical and experimental dependence $\eta(n_w)$ on one piece of graph paper. Indicate the range of applicability of the resulting equation.

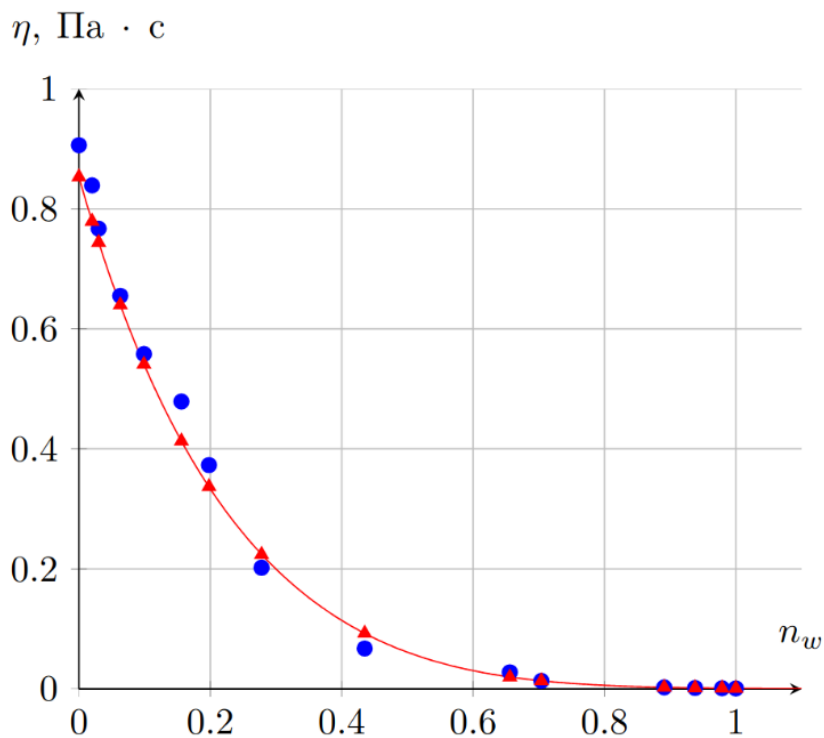
Let us calculate the values using the theoretical dependence $\eta(n_w)$:

$T_g, \text{ K}$	w_g	$\eta, \text{ Pa} \cdot \text{ s}$	n_w	$\ln \eta$	$\ln \frac{\xi \exp(-kn_w)}{\eta} - 1$	$\eta_{\text{th}}, \text{ Pa} \cdot \text{ s}$
195.2	1.000	0.906	0	-0.099	-3.92	0.853
194.8	0.996	0.839	0.020	-0.176	-3.54	0.779
194.6	0.994	0.767	0.030	-0.265	-4.12	0.744
193.9	0.987	0.655	0.063	-0.423	-3.44	0.640
193.1	0.979	0.558	0.099	-0.583	-3.28	0.541
191.8	0.965	0.479	0.156	-0.736	-2.91	0.413
190.8	0.954	0.373	0.198	-0.986	-4.35	0.337
188.6	0.930	0.202	0.278	-1.600	-1.12	0.224
183.4	0.869	0.0674	0.435	-	0.02	0.0926
173.0	0.728	0.0278	0.656	-	-0.12	0.0199
170.0	0.682	0.0134	0.704	-	0.78	0.0135
154.6	0.384	0.0026	0.891	-	1.85	0.0026
149.3	0.252	0.0017	0.938	-	2.10	0.0017
143.8	0.100	0.0012	0.979	-	2.29	0.0012
140.3	0	0.0009	1	-	2.50	0.0010

Let us plot the obtained dependencies $\eta(n_w)$:

By drawing smoothed curves, we conclude:

Answer: The equation is applicable over the entire range ($n_w \in [0, 1]$).



B1

0.40 pts

Using the equation (2), determine the glass transition temperature of the solution.

Let us express w_g :

$$w_g = \frac{m_g}{m_g + m_w} = \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w},$$

then we have:

Answer:

$$T_g = \left[140.3 + 35.272 \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} - 3.879 \left(\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} \right)^2 + 23.467 \left(\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} \right)^3 \right] \text{ K} \approx 162.8 \text{ K}$$

B2

1.50 pts

Propose a linearization to determine the values of the parameters D and T_0 . By plotting a linearized relationship or using the least squares method (OLS), determine the parameter values.

The dependence of $\frac{1}{\ln(\eta/\eta_0)}$ on T is linear with a slope $k = \frac{1}{DT_0}$ and a y-intercept $b = -\frac{1}{D}$. Let us recalculate the data points using this formula and plot the linearized dependence:

$T, ^\circ\text{C}$	$\eta, \text{Pa} \cdot \text{s}$	T, K	$\frac{1}{\ln(\eta/\eta_0)}$
1	0.0201	274	0.1149
4	0.0173	277	0.1169
11	0.0128	284	0.1212
15	0.0104	288	0.1243
35	0.0046	308	0.1383
44	0.0034	317	0.1443
50	0.0031	323	0.1463
56	0.0026	329	0.1501
60	0.0022	333	0.1540
67	0.0020	340	0.1563
71	0.0017	344	0.1604
76	0.00145	349	0.1646
85	0.0013	358	0.1676
90	0.0011	363	0.1724
96	0.0010	369	0.1753
100	0.0009	373	0.1786

Having determined the parameters of the approximating straight line, we obtain a system of equations:

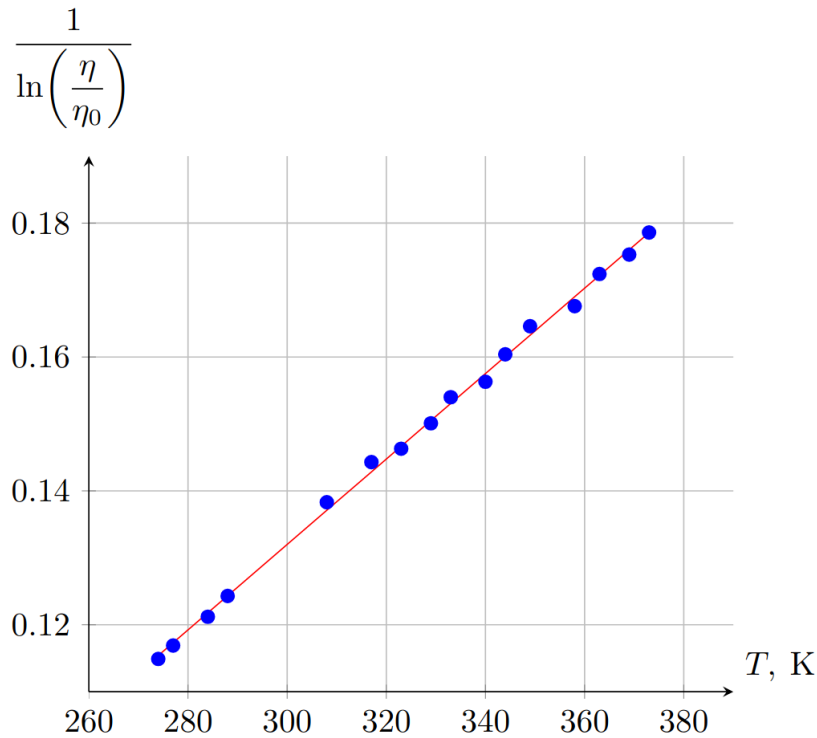
$$\begin{cases} -\frac{1}{D} = -0.059 \\ \frac{1}{DT_0} = 6.38 \cdot 10^{-4} \text{ K}^{-1}, \end{cases}$$

from which it follows:

Answer:

$$T_0 = 92.5 \text{ K}$$

$$D = 16.9$$



C1

0.30 pts

Determine the viscosity of the mixture η_n at the volume fraction of glycerol $c_g = 0.5$ and temperature $T = 25^\circ\text{C}$ using the dependence on the mole fraction of water.

The equation obtained for $\eta_n(n_w)$ is applicable over the entire range. Therefore, it is enough to convert c_g into n_w :

$$n_w = 1 - \frac{w_g \mu_w}{\mu_g - w_g(\mu_g - \mu_w)} = 1 - \frac{\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} \mu_w}{\mu_g - \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} (\mu_g - \mu_w)} \approx 0.80$$

and substitute it into the resulting equation. Thus, we obtain:

Answer: $\eta_n = 6.0 \text{ mPa} \cdot \text{s}$

C2

0.30 pts

Determine the viscosity of the mixture η_t under the same conditions using the temperature dependence.

Similar to the previous section, we substitute the value $T = 25^\circ\text{C} = 298 \text{ K}$ into the equation obtained in part B, from which it follows:

Answer: $\eta_t = 6.8 \text{ mPa} \cdot \text{s}$

C3

0.20 pts

Compare your results. If there are discrepancies, explain the reason for them.

Answer: Considering the large measurement range, the results can be considered consistent. The slight difference is most likely due to inaccurate temperature measurements, since the dependence of viscosity on temperature is very strong.

Restore the missing values v_2 in the table. Use the linearized dependence graph $v_2(\eta)$ for this. Use the empty table columns on the answer sheet to convert the required values.

In the case of a hydrophilic capillary, the dependence of the volumetric flow rate of liquid on viscosity is represented by the equation:

$$Q = \pi r^2 v = \frac{\pi r^4 \Delta P}{8 \eta L} = \frac{\pi r^4 \rho g}{8 \eta},$$

which implies the relation:

$$v = k \frac{\rho}{\eta},$$

where k is a constant value. The density of the mixture can be calculated as:

$$\rho = \rho_g c_g + \rho_w c_w = \rho_w + (\rho_g - \rho_w) c_g$$

Using the expression:

$$n_w = 1 - \frac{\frac{\rho_g c_g}{(\rho_g - \rho_w) c_g + \rho_w} \mu_w}{\mu_g - \frac{\rho_g c_g}{(\rho_g - \rho_w) c_g + \rho_w} (\mu_g - \mu_w)},$$

we express c_g :

$$n_w (\mu_g \rho_w (1 - c_g) + \rho_g c_g \mu_w) = \mu_g \rho_w (1 - c_g)$$

$$c_g = \frac{\mu_g \rho_w (1 - n_w)}{\rho_g \mu_w + \rho_w \mu_g (1 - n_w)}$$

Then the expression for the density of the mixture becomes:

$$\rho = \rho_w + (\rho_g - \rho_w) \frac{\mu_g \rho_w (1 - n_w)}{\rho_g \mu_w + \rho_w \mu_g (1 - n_w)}$$

Using the results of part A, we numerically determine the values of n_w for $\eta < \xi$. Next, we calculate the densities of the mixtures, and for each velocity value in the hydrophilic channel, we calculate the quantity ρ/η and plot the graph $v(\rho/\eta)$:

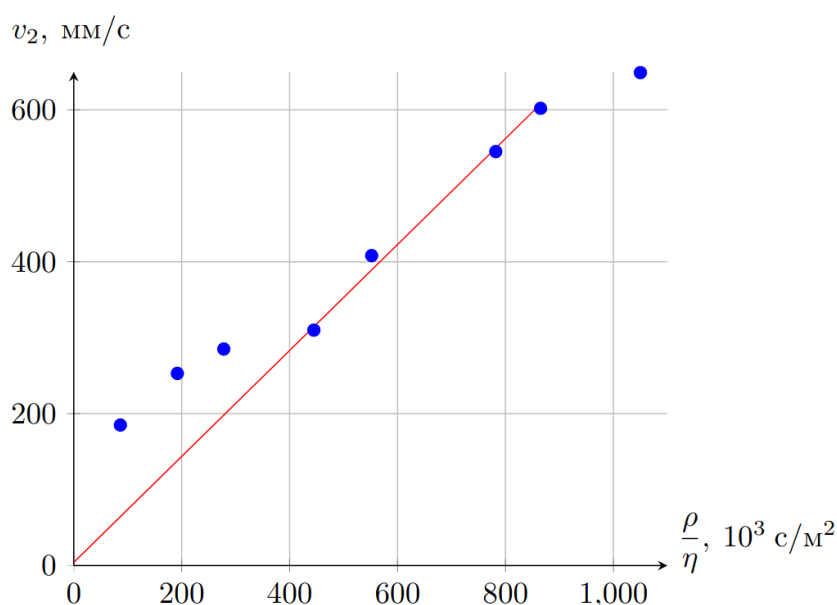
η , mPa · s	v_1 , $\mu\text{m/s}$	v_2 , mm/s	n_w	ρ , $\frac{\text{kg}}{\text{m}^3}$	ρ/η , $\frac{\text{s}}{\text{m}^2} \cdot 10^3$
0.95	4.25	649	1.000	1000	1050
1.18	4.86	602	0.978	1021	865
1.32	5.03	545	0.966	1032	782
1.92	8.03	408	0.926	1060	552
2.41	9.80	310	0.905	1072	445
3.95	15.2	285	0.847	1100	278
5.81	18.2	253	0.804	1115	192
13.2	24.3	185	0.707	1141	86.4
18.4	25.5	-	0.666	1150	62.5
24.8	29.3	-	0.627	1157	46.7
72.3	38.8	-	0.475	1177	16.3
78.0	40.3	-	0.463	1178	15.1
224	51.0	-	0.278	1194	5.33
515	55.3	-	0.110	1204	2.34
989	59.4	-	-	1260	1.27
1400	65.2	-	-	1260	0.900

The straight line must pass through the origin (0,0). By fitting it to the four initial data points, we find k :

$$k = 0.674 \cdot 10^{-6} \frac{\text{m}^3}{\text{s}^2}$$

Now knowing k , we can restore the dependence $v_2(\eta)$:

η , mPa · s	v_1 , $\mu\text{m/s}$	v_2 , mm/s	n_w	ρ , $\frac{\text{kg}}{\text{m}^3}$	ρ/η , $\frac{\text{s}}{\text{m}^2} \cdot 10^3$
0.95	42.5	649	1.000	1000	1050
1.18	48.6	602	0.978	1021	865
1.32	50.3	545	0.966	1032	782
1.92	80.3	408	0.926	1060	552
2.41	98.0	310	0.905	1072	445
3.95	152	285	0.847	1100	278
5.81	182	253	0.804	1115	192
13.2	243	185	0.707	1141	86.4
18.4	255	42.2	0.666	1150	62.5
24.8	293	31.5	0.627	1157	46.7
72.3	388	11.0	0.475	1177	16.3
78.0	403	10.2	0.463	1178	15.1
224	510	3.59	0.278	1194	5.33
515	553	1.58	0.110	1204	2.34
989	594	0.856	-	1260	1.27
1400	652	0.607	-	1260	0.900



D2

1.50 pts

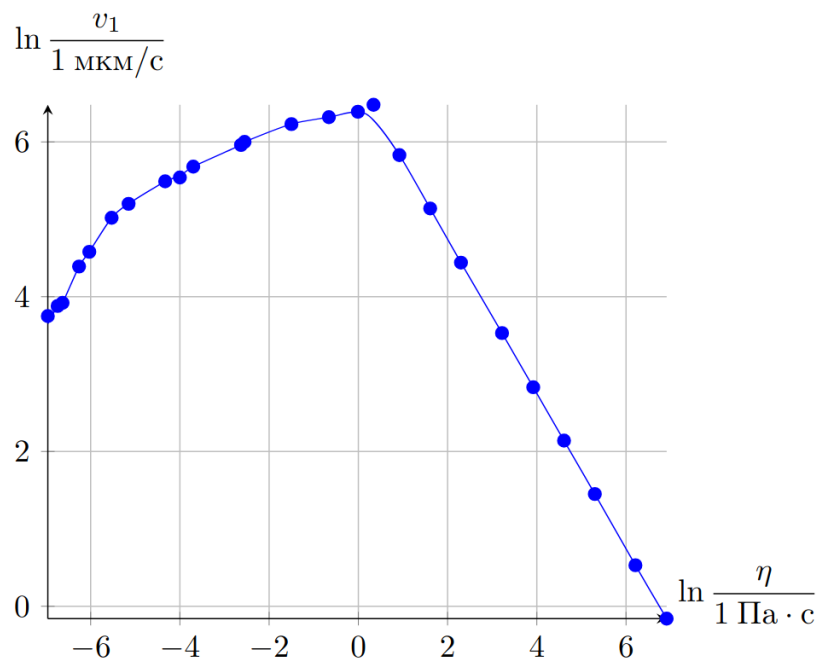
Construct a graph of $v_1(\eta)$ in the range from $\eta_1 = 10^{-3}$ Pa · s to $\eta_2 = 10^3$ Pa · s in the coordinates that are most convenient for graphical analysis. All points plotted on the graph must be tabulated on the answer sheet.

At high viscosities, the velocities in hydrophobic and hydrophilic capillaries are equal. At $\eta = 1.400$ Pa · s, $v_1 > v_2$; therefore, we assume that $v_1 = v_2$ for $\eta > \eta_{\text{max}} = 1.400$ Pa · s. Let us enter the points into the table and construct a graph in logarithmic coordinates, since the values of the quantities under study vary by several orders of magnitude.

η , mPa · s	v_1 , $\mu\text{m/s}$	v_2 , mm/s	n_w	ρ , $\frac{\text{kg}}{\text{m}^3}$	ρ/η , $\frac{\text{s}}{\text{m}^2} \cdot 10^3$	$\ln \frac{v_2}{1 \text{ mm/s}}$	$\ln \frac{\eta}{1 \text{ Pa} \cdot \text{s}}$
0.95	42.5	649	1.000	1000	1050	3.75	-6.96
1.18	48.6	602	0.978	1021	865	3.88	-6.74
1.32	50.3	545	0.966	1032	782	3.92	-6.63
1.92	80.3	408	0.926	1060	552	4.39	-6.26
2.41	98.0	310	0.905	1072	445	4.58	-6.03
3.95	152	285	0.847	1100	278	5.02	-5.53
5.81	182	253	0.804	1115	192	5.20	-5.15
13.2	243	185	0.707	1141	86.4	5.49	-4.33
18.4	255	42.2	0.666	1150	62.5	5.54	-4.00
24.8	293	31.5	0.627	1157	46.7	5.68	-3.70
72.3	388	11.0	0.475	1177	16.3	5.96	-2.63
78.0	403	10.2	0.463	1178	15.1	6.00	-2.55
224	510	3.59	0.278	1194	5.33	6.23	-1.50
515	553	1.58	0.110	1204	2.34	6.32	-0.66
989	594	0.856	-	1260	1.27	6.39	-0.01
1400	652	0.607	-	1260	0.900	6.48	0.34

η , Pa · s	v_1 , $\mu\text{m/s}$	v_2 , mm/s	n_w	ρ , $\frac{\text{kg}}{\text{m}^3}$	ρ/η , $\frac{\text{s}}{\text{m}^2}$	$\ln \frac{v_1}{1 \mu\text{m/s}}$	$\ln \frac{\eta}{1 \text{ Pa} \cdot \text{s}}$
2.5	340	-	-	1260	504	5.83	0.92
5.0	170	-	-	1260	252	5.14	1.61
10.0	84.9	-	-	1260	126	4.44	2.30
25.0	34.0	-	-	1260	50.4	3.53	3.22
50.0	17.0	-	-	1260	25.2	2.83	3.92
100	8.49	-	-	1260	12.6	2.14	4.61
200	4.25	-	-	1260	6.30	1.45	5.30
500	1.70	-	-	1260	2.52	0.53	6.21
1000	0.85	-	-	1260	1.26	-0.16	6.91

Let us plot the obtained points on the graph: $\ln \left(\frac{v_1}{1 \mu\text{m/s}} \right)$ versus $\ln \left(\frac{\eta}{1 \text{ Pa} \cdot \text{s}} \right)$.



D3

0.30 pts

Determine the value of the mixture viscosity η_{max} such that the value $v_2(\eta_{max})$ is maximum.

Answer:

$$\eta_{max} = 1.400 \text{ Pa} \cdot \text{s}$$